



VIVEKANAND EDUCATION SOCIETY

INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

(Affiliated to University of Mumbai, Approved by AICTE & Recognized by Govt. of Maharashtra)

Smart Fitness Activity Prediction System using ML & DL Machine Learning & Deep Learning

in Semester - VI by

Himanshi Bajaj - 26

under the guidance of

Mrs.Dipti Karani



Department of Information Technology

**Vivekanand Education Society's Institute of
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ABSTRACT

In today's digital world, fitness and health monitoring have become very important. Many people use wearable devices such as fitness bands, smartwatches, and mobile applications to track their daily activities. These devices generate a large amount of data such as number of steps, calories burned, and heart rate.

However, this data is often not properly analyzed. Most users only see raw numbers and do not understand what it actually means for their health. There is a need for an intelligent system that can analyze this data and provide meaningful insights.

The main aim of this project is to develop a Smart Fitness Activity Prediction System using Machine Learning (ML) and Deep Learning (DL) techniques. The system analyzes fitness data and predicts the activity level of a user (low, medium, high).

Different ML algorithms such as Linear Regression, Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and K-Means Clustering are used. In addition, a Deep Learning model using Artificial Neural Networks (ANN) is also implemented.

The system also includes data visualization using graphs, which helps in understanding patterns easily. An interactive interface is developed using Streamlit so that users can input their data and get predictions instantly.

This project demonstrates how Artificial Intelligence can be used in real-life applications like health monitoring and fitness tracking.



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INTRODUCTION

Technology has changed the way people monitor their health and fitness. Nowadays, many individuals use smart devices like fitness trackers and mobile applications to track their daily physical activities. These devices collect various types of data such as steps walked, calories burned, and heart rate.

Even though this data is available, most users do not have the tools or knowledge to analyze it. They cannot understand patterns or predict their fitness level. This leads to underutilization of valuable data.

Machine Learning (ML) and Deep Learning (DL) are powerful techniques that can analyze data, find patterns, and make predictions. These techniques are widely used in healthcare, finance, and many other domains.

In this project, ML and DL techniques are applied to fitness data to predict the activity level of a user. The system not only predicts results but also provides visual insights using graphs and charts.

This makes the system useful, user-friendly, and easy to understand. It helps users gain better knowledge about their fitness and make informed decisions.

PROBLEM STATEMENT

In today's world, people are generating a large amount of fitness data using wearable devices and mobile applications. However, this data is not properly analyzed or utilized.

The main problems are:

- Users cannot understand their fitness data clearly
- Data is available but not converted into useful information
- No simple system exists to predict activity levels
- Lack of intelligent tools for health analysis

Because of this, users are unable to track their progress effectively or improve their health.

Therefore, the problem is to design and develop a system that can:

- Analyze fitness data
- Identify patterns and trends
- Predict user activity level
- Provide insights using graphs

The solution should be simple, interactive, and easy to use.



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OBJECTIVES

The main objectives of this project are:

- To collect and analyze fitness data such as steps, calories, and heart rate
- To apply different Machine Learning algorithms for prediction
- To classify user activity levels into low, medium, and high
- To implement Deep Learning (ANN) for improved performance
- To compare different models based on accuracy
- To visualize data using graphs and charts
- To identify patterns and relationships in the dataset
- To build an interactive system using Streamlit
- To make the system easy to understand and user-friendly

These objectives help in building a complete and meaningful project.



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DATASET DESCRIPTION

The dataset used in this project is a structured fitness dataset. It contains information related to user activity.

Features of Dataset:

1. **Steps**
Represents the number of steps taken by a user in a day.
2. **Calories**
Indicates the number of calories burned.
3. **Heart Rate**
Represents the pulse rate of the user.

Target Variable:

- **Activity Level**
 - 0 → Low Activity
 - 1 → Medium Activity
 - 2 → High Activity

Dataset Characteristics:

- Small dataset
- Numeric values
- Clean and structured data
- Suitable for ML algorithms

The dataset is designed in such a way that it shows relationships between physical activity and health parameters.



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METHODOLOGY / WORKFLOW

The project follows a step-by-step process:

Step 1: Data Collection

A sample dataset is created manually for demonstration purposes.

Step 2: Data Preprocessing

The dataset is organized into input variables (features) and output variable (target). Data is checked for consistency.

Step 3: Data Visualization

Graphs such as scatter plots and histograms are used to understand patterns in data.

Step 4: Model Implementation

Different ML algorithms and ANN model are applied.

Step 5: Model Training

Models are trained using the dataset.

Step 6: Model Evaluation

Accuracy and confusion matrix are used to evaluate performance.

Step 7: Interface Development

A web-based interface is created using Streamlit.



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ALGORITHMS USED

1. Linear Regression

Used to predict calories based on steps. It shows the relationship between two variables.

2. Logistic Regression

Used for classification problems. It predicts the probability of a class.

3. Decision Tree

A tree-based model that makes decisions using rules.

4. Random Forest

An advanced model that uses multiple decision trees to improve accuracy.

5. K-Nearest Neighbors (KNN)

Predicts output based on nearest data points.

6. K-Means Clustering

Groups similar data into clusters.

7. Artificial Neural Network (ANN)

A deep learning model inspired by the human brain.



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MATHEMATICAL FORMULATION

Linear Regression

$$y = mx + c$$

Used to find a linear relationship.

Logistic Regression

$$P = 1 / (1 + e^{-x})$$

Used for classification.

KNN Distance Formula

$$\text{Distance} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

ANN Formula

$$\text{Output} = \text{Activation}(WX + b)$$

Loss Function

Cross-entropy is used to measure error.

These formulas help the model learn patterns from data

PERFORMANCE ANALYSIS

The models are evaluated using accuracy and confusion matrix.

Observations:

- Random Forest gives highest accuracy
- ANN performs well with training
- KNN works well for small dataset
- Decision Tree is easy to understand

Confusion matrix helps in identifying correct and incorrect predictions.

Overall, the models show good performance in predicting activity levels.



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SYSTEM INTERFACE

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★ Problem Statement ▼



Smart Fitness Activity Prediction System

Problem Statement

In today's digital world, fitness apps and wearable devices generate large amounts of user data such as steps, calories burned, and heart rate. However, this data is often not analyzed effectively, and users fail to gain meaningful insights about their health and activity levels. This project aims to build an intelligent system that can:

- Analyze fitness data
- Identify patterns
- Predict user activity levels using Machine Learning and Deep Learning models

Why This Problem? ☞

- Increasing use of fitness trackers
- Lack of intelligent analysis tools
- Need for personalized health insights

Observations

- Higher steps generally lead to higher calorie burn
- Increased heart rate indicates higher physical activity
- Users with similar patterns can be grouped together

What Makes This Project Different?

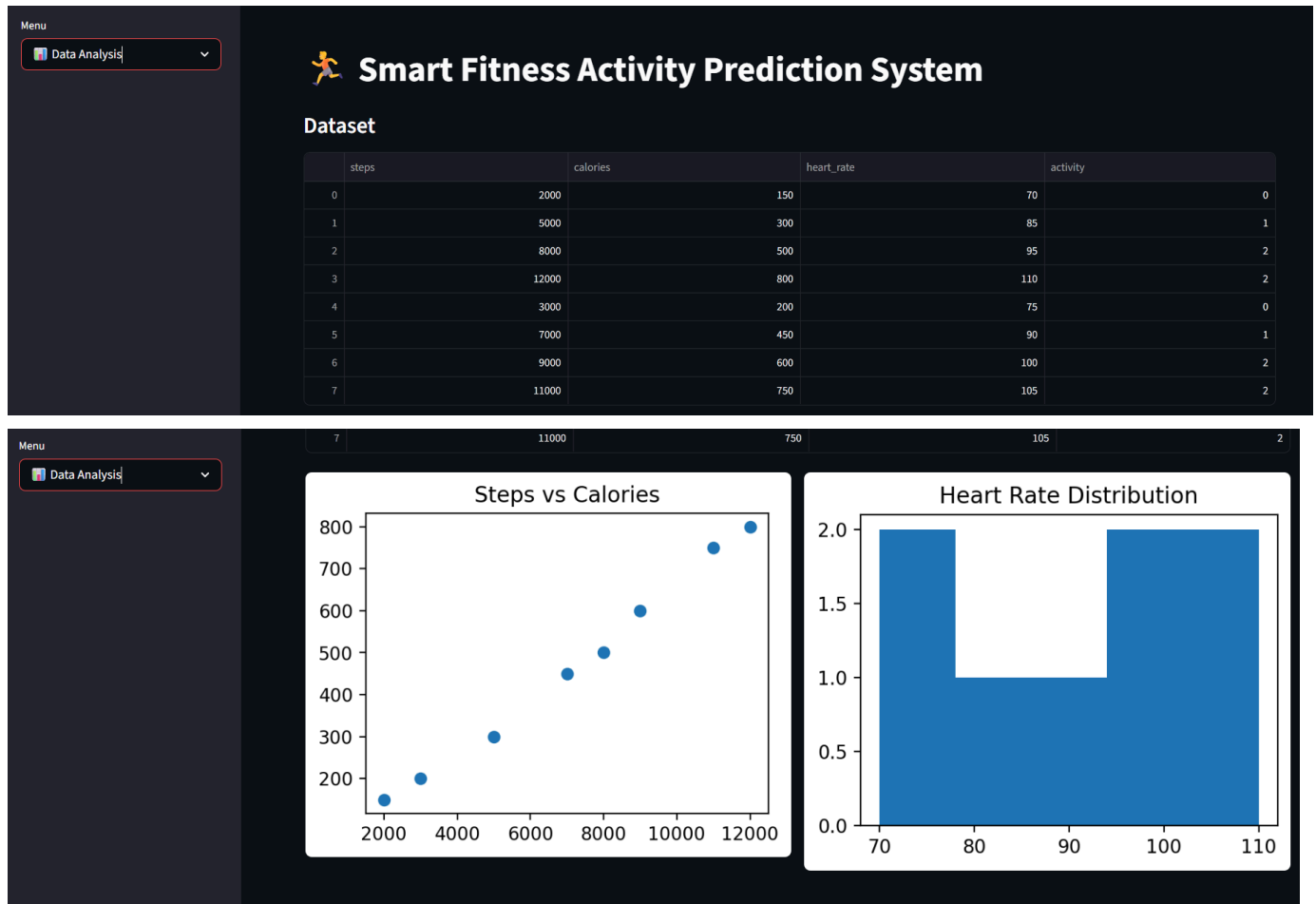
- Combines ML + DL in one system
- Uses multiple algorithms for comparison
- Provides visual insights using graphs
- Interactive prediction system

PROBLEM STATEMENT



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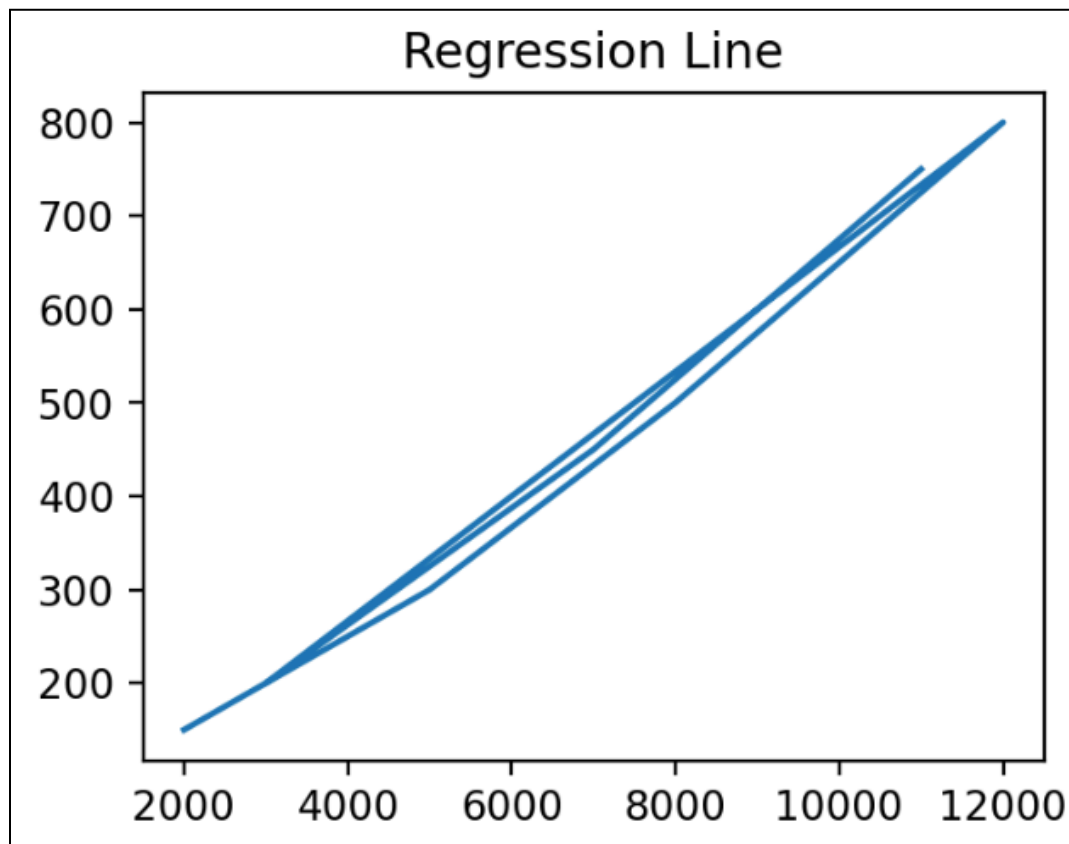
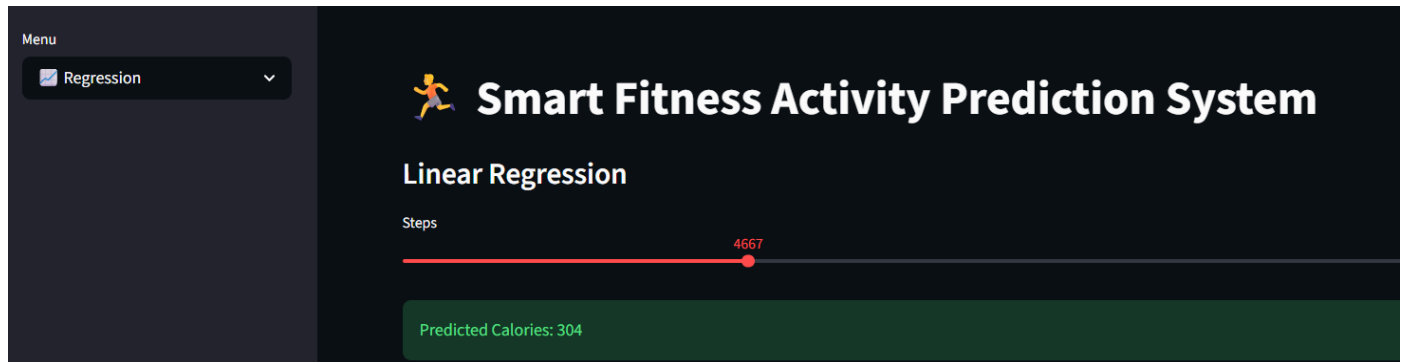


DATA ANALYSIS



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REGRESSION



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Classification



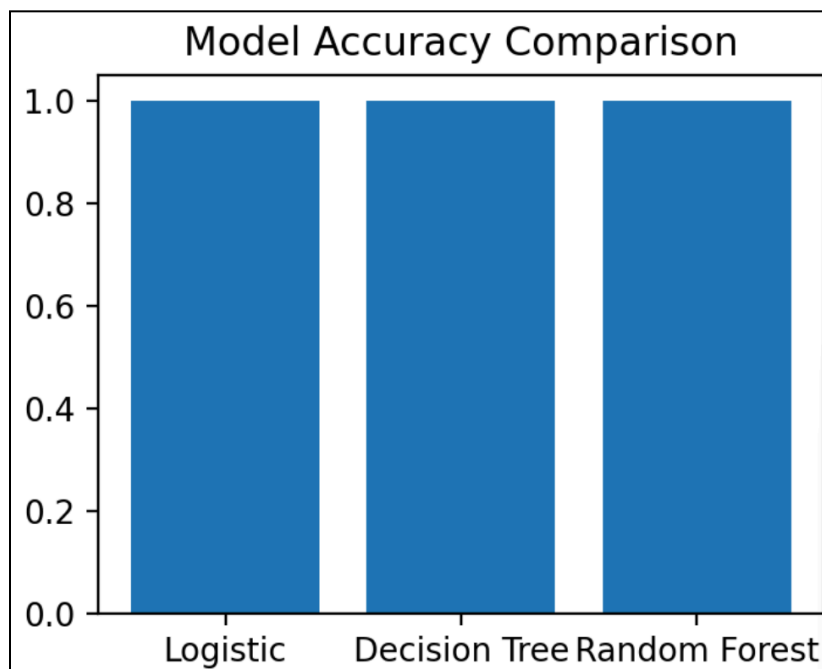
Smart Fitness Activity Prediction System

Classification Models

Logistic Accuracy: 1.0

Decision Tree Accuracy: 1.0

Random Forest Accuracy: 1.0



CLASSIFICATION



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KNN

 **Smart Fitness Activity Prediction System**

KNN Prediction

Steps

6000

Calories

400

Heart Rate

85

Predicted Activity Level: 1

KNN

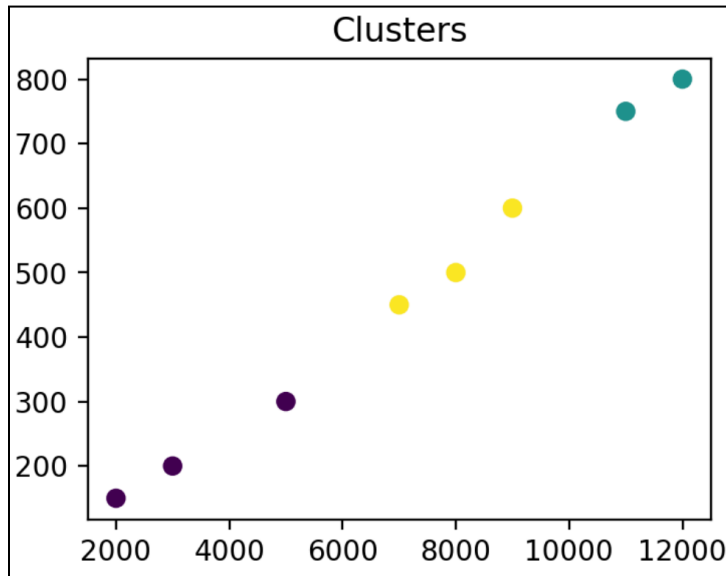
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Clustering

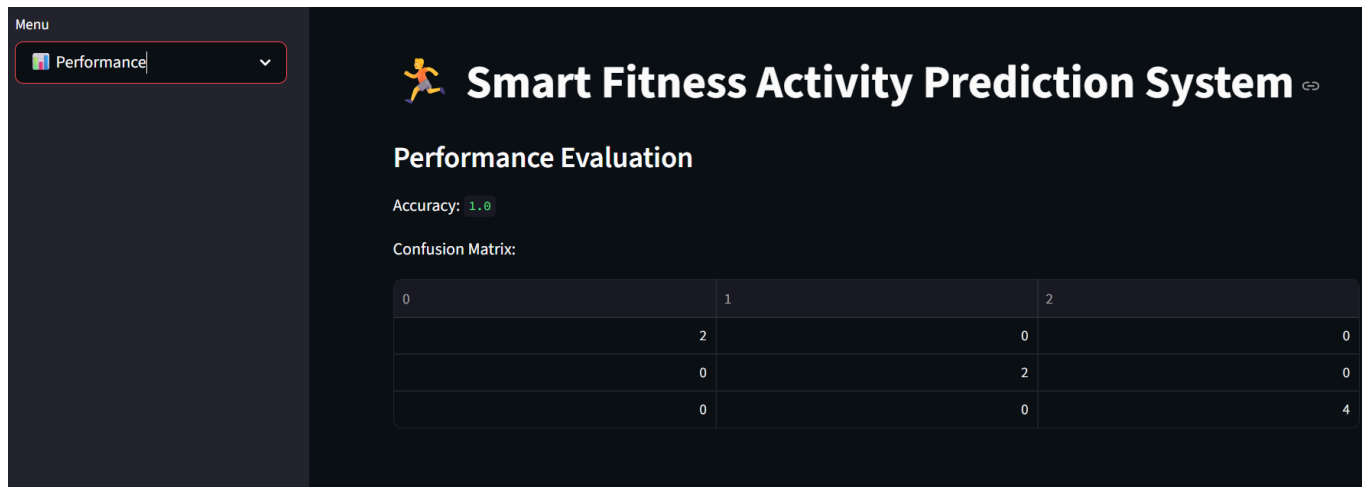
 **Smart Fitness Activity Prediction System**

K-Means Clustering

	steps	calories	heart_rate	activity	cluster	
0	2000	150	70	0	0	
1	5000	300	85	1	0	
2	8000	500	95	2	2	
3	12000	800	110	2	1	
4	3000	200	75	0	0	
5	7000	450	90	1	2	
6	9000	600	100	2	2	
7	11000	750	105	2	1	



CLUSTERING



PERFORMANCE - CONFUSION MATRIX



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★ Conclusion ▼



Smart Fitness Activity Prediction System

Conclusion

This project demonstrates how Machine Learning and Deep Learning techniques can be applied to fitness data analysis.

Key Results:

- Random Forest achieved high accuracy
- ANN provided strong predictive capability
- Clustering revealed user activity patterns

This system can be further extended into real-time health monitoring applications.

CONCLUSION



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HYPERPARAMETER TUNING

Hyperparameters are adjusted to improve model performance.

Examples:

- KNN → number of neighbors ($k = 3$)
- Random Forest → number of trees
- ANN:
 - Epochs = 50
 - Hidden neurons = 8
 - Activation function = ReLU

Tuning helps in achieving better accuracy and reducing errors.

LIMITATIONS

- Dataset is small
- Data is not real-time
- Limited features
- Results may not be highly accurate
- Model performance can improve with more data

OBSERVATIONS

- More steps lead to more calories burned
- Higher heart rate indicates higher activity
- Data can be grouped using clustering
- ML models can identify patterns easily



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APPLICATIONS

- Fitness tracking applications
- Health monitoring systems
- Smart wearable devices
- Personalized workout plans
- Medical data analysis

CONCLUSION

This project successfully demonstrates how Machine Learning and Deep Learning techniques can be applied to fitness data analysis.

Different algorithms were implemented and compared. Random Forest and ANN performed best. The system provides accurate predictions and useful insights.

In the future, this project can be improved by using real-time data, larger datasets, and more advanced models.

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